

Methodological Guide to Shared Embedding Spaces

Combining Hub-Dimension, Structural Similarity, and Platonic Representation Tests

A practical methodology for testing direction, geometry, topology, shared spaces, and exportability between encoders

Running example used throughout

Image item: pixels showing a cat sitting on a chair.

Text item: "A cat is sitting on a chair."

The methodological question is whether different encoders - image and text - preserve enough of the same relational structure that their vectors can be compared, mapped, or routed through a shared Hub.

What this document combines

This guide synthesizes two prior documents into a methodological framework. The first document explains HUB_DIM, reconstruction, SVD slicing, R@1 versus verification AUC, anisotropy, hubness, and collapse. The second document explains the Observer/Platonic framing, local topology, global geometry, CKA, CCA, Procrustes, directional mapping, and the 7-space experiment. The result here is not a philosophical essay and not a raw PDF merge. It is a reusable methodology for deciding whether embedding spaces share structure, at what scale, and whether that structure can be exported from one model space into another.

Core claim: models do not need to produce identical vectors. They may instead produce different coordinate charts over a partially shared relational manifold. The methodology asks which invariants survive: local neighbors, distance rankings, global geometry, linear subspaces, near-isometry, or retrieval-level pair identity.

Terminology and Setup

A representation matrix stores one vector per aligned item. If encoder A produces d_A features for N items, then A has shape $N \times d_A$. Row i is the representation of item i. In this project, row i may correspond to the same COCO image-caption item across image encoders and text encoders.

Object	Meaning	Cat-chair example
Item i	The underlying semantic unit shared across modalities.	The same scene: cat sitting on a chair.
Image representation	Vector produced from pixels.	DINOv2 encodes the image pixels.
Text representation	Vector produced from a caption.	BGE/GPT-2 encodes: "A cat is sitting on a chair."
Space Z_A	The coordinate system used by encoder A.	Image embedding space.
Space Z_B	The coordinate system used by encoder B.	Text embedding space.
Shared Hub H	A learned common coordinate system used as a bridge.	The cat item receives a 512-d Hub coordinate.

The central distinction

Same item does not mean same raw input.

Image arm input: pixels of the cat-on-chair image.

Text arm input: "A cat is sitting on a chair."
Same row means same referent, not same modality.

This alignment is what makes the experiment powerful: for every row i , all representations describe the same underlying semantic item, even if each encoder observes a different projection of it.

Method 1 - Local Topology: kNN Neighborhood Overlap

Question tested: Do two encoders agree on which items are near which other items? This is the weakest but often most robust form of shared structure.

For every item i :
 $N_A^k(i)$ = k nearest neighbors of item i in space A
 $N_B^k(i)$ = k nearest neighbors of item i in space B
 $\text{Overlap}_k(i) = |N_A^k(i) \cap N_B^k(i)| / k$

Cat-chair example

In image space, the cat-on-chair item may be near other indoor-cat images, sofa-chair scenes, and pet-on-furniture images. In text space, the sentence "A cat is sitting on a chair." may be near captions such as "a cat rests on a sofa" or "a kitten sits on furniture." If the neighbor lists overlap, the spaces share local topology even if the axes and distances are different.

Interpretation: high kNN overlap suggests that the encoders preserve neighborhood topology. Low global alignment with high local overlap suggests that the spaces may be nonlinear warps of the same manifold rather than unrelated spaces.

Method 2 - Scale of Agreement: k Sweep

Question tested: Is the agreement only immediate/local, or does it survive at larger scales? Instead of using one k , measure a curve over k values.

Recommended k values: 1, 5, 10, 20, 50, 100, 500
Report both raw overlap and overlap/chance.

Cat-chair example

At $k=5$, both spaces may agree on very similar cat-on-furniture examples. At $k=100$, one space may group by object category (cat), while another may group by scene (indoor furniture). If overlap/chance falls as k grows, the agreement is local rather than global.

Methodological rule: raw overlap naturally rises with k because chance also rises. Therefore the ratio overlap/chance is essential for deciding whether agreement remains meaningful at larger scales.

Method 3 - Shared Distance Ranking

Question tested: Do two spaces rank distances similarly, even if the exact distances differ? This is stronger than neighbor overlap and weaker than full metric identity.

For item i , compare rankings of distances from i to all other items:
 $\text{rank}_A(d_A(i,j))$ versus $\text{rank}_B(d_B(i,j))$

Use Spearman correlation or rank-biased overlap.

Cat-chair example

If both image and text spaces rank "cat on sofa" closer than "airplane in sky" and closer than "person riding bicycle," they share relational ordering around the cat-chair item. The exact numeric distances do not need to match.

Interpretation: high ranking agreement means the relational geography around points is similar. This supports a shared manifold view without requiring a single global coordinate system.

Method 4 - Global Geometry: CKA / RSA

Question tested: Do two representation spaces have globally similar relational fingerprints? CKA compares the pattern of similarities among all items and is useful even when dimensions differ.

$$CKA(X, Y) = \frac{||X^T Y||_F^2}{(||X^T X||_F ||Y^T Y||_F)}$$

Use CKA as a symmetric structural map: A vs B = B vs A.

Cat-chair example

CKA does not ask whether the cat sentence vector equals the cat image vector. It asks whether the full relational pattern - cat items near cat items, furniture scenes near furniture scenes, vehicles far from cats - is similarly organized across the two matrices.

Interpretation: high CKA is evidence of global structural similarity, but it does not prove that one embedding can be exported into the other. For exportability, add a directional mapping test.

Method 5 - Directional Mapping: A -> B Is Not the Same as B -> A

Question tested: Can a vector from source space A be transformed into target space B well enough to retrieve its true counterpart? Direction matters because information content and geometry may be asymmetric.

Learn on training rows only:

$B_{\text{hat}} = A W$

Evaluate on held-out rows:

$B_{\text{hat_eval}} = A_{\text{eval}} W$

Then retrieve the nearest B_{eval} vector for each predicted $B_{\text{hat_eval}}$.

Direction	Meaning	Example
image -> text	Can image vectors be exported into a text embedding space?	Pixels of cat-on-chair -> bge vector for "A cat is sitting on a chair."
text -> image	Can text vectors predict image embedding targets?	Caption vector -> image vector of the cat scene.
small image -> large image	Can a compact encoder recover the richer encoder space?	DINO-small -> DINO-large.
large image -> small image	Can the richer encoder collapse down to the compact space?	DINO-large -> DINO-small.

With 7 spaces there are 21 symmetric pairs but 42 directed mappings. A strong A -> B and weak B -> A result is not a contradiction; it may reveal information containment or target-space complexity.

Method 6 - Retrieval Metrics: R@1, R@5, R@10

Question tested: After mapping, does the predicted vector retrieve the exact paired target from many candidates? This is the strictest operational test.

```
For item i:  
query = mapped source vector A_i  
candidates = all target vectors B_1, ..., B_n  
R@1 succeeds if  $\text{argmax}_j \text{sim}(\text{query}, B_j) = i$   
R@5 succeeds if the true i is among the top 5 candidates.
```

Cat-chair example

After mapping the image vector of the cat-on-chair scene into text space, the nearest text target should be the caption vector for "A cat is sitting on a chair." R@1 asks for exact top-1 recovery; R@5 allows the correct caption to appear among the top five.

Interpretation: R@1 is harsh because a few false candidates can beat the true pair. R@5 and R@10 are useful when the downstream UI displays a short results grid.

Method 7 - Verification AUC: Is There Coarse Correspondence?

Question tested: Are true pairs generally scored higher than random pairs? This is weaker than retrieval but useful for diagnosing whether signal remains after R@1 collapses.

```
Verification asks whether:  
 $\text{sim}(A_i, B_i) > \text{sim}(A_i, B_j)$ , where  $j \neq i$   
 $\text{AUC} = P(\text{score}_{\text{true}} > \text{score}_{\text{random}})$ 
```

Cat-chair example

The true cat-chair image/text pair may score 0.80, while most random pairs score 0.30-0.60. AUC will be high. But if two wrong captions score 0.83 and 0.82, R@1 fails even though coarse correspondence remains.

Methodological reading: AUC high with R@1 low means the pipeline still preserves broad correspondence but not enough pair-specific discriminability for exact identification.

Method 8 - PCA + Orthogonal Procrustes: Clean Near-Isometry Test

Question tested: Are the spaces nearly the same geometry with different coordinates? Procrustes is deliberately constrained: it can rotate or reflect but cannot arbitrarily rescale or invent information.

```
Recommended clean funnel:  
A -> PCA_256  
B -> PCA_256  
Find R such that  $Y \approx X R$  and  $R^T R = I$   
Evaluate by retrieval on held-out items.
```

Cat-chair example

If the cat-chair image vector and its text vector occupy corresponding positions after both spaces are reduced to 256 dimensions, a rotation/reflection should align them. Success means the two spaces are close to the same geometry in different coordinates.

Interpretation: strong Procrustes indicates near-isometry. Weak Procrustes but strong Ridge means the spaces share linear structure but not pure rotation/reflection. Strong kNN but weak Procrustes suggests shared topology with nonlinear warping.

Method 9 - Ridge / Linear Stitching: Exportability Without Neural Training

Question tested: Can a single linear map translate between spaces? Ridge is expressive enough to undo rotations, permutations, scalings, and anisotropy, but limited enough that success remains meaningful.

```
Fit W by ridge regression on paired training rows:  
min_W ||A_train W - B_train||^2 + lambda ||W||^2  
Then test on held-out rows only.
```

Cat-chair example

Fit W using many image-caption pairs. On a held-out cat-chair item, use the image vector and W to predict the text-space vector. If the predicted vector retrieves the caption "A cat is sitting on a chair," the representation has been exported.

Methodological rule: do not tune on the evaluation rows. Use train data for fitting, validation if needed for hyperparameters, and held-out evaluation only once for final R@K.

Method 10 - CCA: Shared Linear Latent Subspace

Question tested: Do the two spaces share correlated directions even if each also contains private information?

```
CCA finds directions a and b such that:  
u = X a  
v = Y b  
corr(u, v) is maximized  
A long sequence of high canonical correlations suggests a shared linear latent subspace.
```

Cat-chair example

One CCA direction may encode animal-vs-non-animal, another indoor-vs-outdoor, another furniture context. The exact vector coordinates differ, but the same semantic factors can be recovered as correlated directions.

Interpretation: CCA high with CKA moderate often means there is a shared subspace plus private, modality-specific information.

Method 11 - Shared Hub: Building a Common Coordinate System

Question tested: Can multiple spaces be routed into one common coordinate system that preserves enough cross-space structure for retrieval or translation?

```
For each aligned training item i:  
z_i = [x_small | x_base | x_large | x_bge | x_gpt2]  
Z has shape N x total_width
```

Apply PCA/SVD and keep HUB_DIM components, e.g. 512.
The result is H in $\mathbb{R}^{(N \times \text{HUB_DIM})}$.

Cat-chair example

The cat-chair item contributes one DINO-small image vector, one DINO-base image vector, one DINO-large image vector, one bge caption vector for "A cat is sitting on a chair," and one GPT-2 caption vector. Their standardized concatenation helps define a shared Hub coordinate for that semantic item.

At runtime, production does not need the full concatenation. Once the Hub is defined, each encoder learns its own encoder-to-Hub map: $X_k W_k \approx H$. A new image can enter the Hub using only one encoder.

Method 12 - Choosing HUB_DIM by Sweep, Not Intuition

Question tested: What Hub width preserves information without making the mapping underpowered? HUB_DIM is a trade-off between compression and statistical support.

Choice	Benefit	Risk
Hub too narrow	Easy to fit; fewer parameters.	Bottleneck; loses independent directions; poor reconstruction.
Hub too wide	Can preserve more information.	Too few rows per dimension; mapping may become underpowered.
Measured sweet spot	Largest useful dimension supported by data.	Must be chosen by sweep, not by argument.

Example sweep:

HUB_DIM in {256, 512, 768, 1024, 1536, 2048}

Measure: reconstruction/retention, rows per dimension, transfer performance, R@K.

Cat-chair example

If the Hub is too narrow, subtle differences between "cat on chair," "cat on sofa," and "dog on chair" may collapse. If the Hub is too wide relative to the number of examples, the learned maps may fit noise and fail on held-out cat-chair examples.

Methodological rule: compute SVD once, then slice prefixes of singular components for different Hub widths. This makes the sweep cheap and evidence-based.

Method 13 - Directions, Geometry, Topology, and Shared Space

Concept	Methodological meaning	Cat-chair example
Direction	A directed export path, such as $A \rightarrow B$.	Image vector $\rightarrow$ text vector is not identical to text vector $\rightarrow$ image vector.
Topology	Who is near whom; local adjacency.	Cat-chair is near cat-sofa and kitten-chair in both spaces.
Geometry	Distances, angles, scale, and global metric structure.	Both spaces place cats near cats and far from airplanes with similar distance patterns.
Shared subspace	Some directions encode common factors.	Animal, furniture, indoor-scene directions are recoverable in both spaces.
Shared Hub	A learned bridge coordinate system.	The item receives one 512-d Hub vector used by multiple encoders.
Near-isometry	Same geometry up to rotation/reflection.	PCA-Procrustes aligns the cat-chair neighborhoods with minimal distortion.

Method 14 - Anisotropy, Hubness, and Collapse Diagnostics

Question tested: Did retrieval fail because there is no shared structure, or because the measurement pipeline lost discriminative geometry?

Failure mode	Meaning	Example symptom
Anisotropy	Vectors concentrate in a narrow cone or share a strong common component.	Many captions have cosine near 0.99 even when semantically different.
Hubness	A few candidate vectors become nearest neighbors for many queries.	Many different image queries retrieve the same few text vectors.
Arm collapse	One side loses pair-specific discrimination.	True pair is above random pairs, but not top-1 among all candidates.
Hub-induced collapse	Projection or bottleneck destroys distinguishing directions.	After Hub reconstruction, cat-chair and cat-sofa become too similar.

Cat-chair example

Before collapse: true cat-chair similarity = 0.85, random dog/car captions = 0.20. After collapse: true cat-chair = 0.92, random captions = 0.88. The true cosine increased, but the margin vanished; retrieval becomes worse.

Methodological rule: collapse in R@1 alone does not prove absence of shared structure. If AUC remains high, the pipeline may preserve coarse correspondence while losing retrieval-level precision.

Method 15 - Model-Induced vs Universal/Platonic Explanations

Question tested: Is similarity explained by shared model properties, or by the fact that both models discover structure in the same world?

Hypothesis	Prediction	Cat-chair example
Model-induced similarity	High agreement appears mainly within the same modality, architecture, objective, or model family.	DINO-small and DINO-base agree strongly, but image-text agreement is weak.
Universal relational structure	Some agreement survives across modality and architecture boundaries.	Image encoder and text encoder both place cat-chair near other cat/furniture items.
Mixed explanation	Similarity = world-induced component + model/training-induced component.	Local cat-neighborhoods agree, but global geometry differs by modality.

Use metadata for each of the 21 pairs: same/different modality, architecture family, training objective, and model family. Compare average kNN, rank agreement, CKA, and R@K across those groups. With only 21 pairs this is exploratory, but it can guide model expansion.

Recommended Experimental Order

Step	Test	Why first/next	Output
1	kNN overlap over k sweep	Cheapest; detects local topology.	21-pair local topology matrix.
2	CKA / RSA	Global structural comparison without mapping.	21-pair geometry matrix.
3	PCA-256 + Procrustes	Clean near-isometry test.	42 directional R@K results.
4	Ridge mapping	Tests linear exportability.	42 directional R@1/R@5/R@10.
5	CCA	Finds shared subspaces.	Canonical correlation

			spectrum.
6	Hub sweep	Chooses HUB_DIM empirically.	Retention, rows/dim, transfer, R@K.
7	Anisotropy/hubness diagnostics	Explains collapse modes.	Effective rank, mean cosine, hub count distribution.

Decision Rules

Observed pattern	Methodological interpretation
kNN high + CKA low	Shared local topology, different global geometry.
CKA high + Procrustes high	Nearly same geometry, different coordinates.
CKA high + Ridge high + Procrustes low	Shared structure is linearly exportable but not pure rotation/reflection.
CCA high + CKA moderate	Shared latent subspace plus private information.
A -> B strong, B -> A weak	Directional information asymmetry or easier target geometry.
AUC high + R@1 low	Coarse correspondence survives; pair-specific resolution is insufficient.
R@K collapse + high anisotropy/hubness	Measurement or representation collapse may obscure real structure.
Only within-family pairs high	Model-induced similarity is the stronger explanation.
Cross-modal, cross-architecture agreement remains high	World-induced relational structure becomes more plausible.

Minimal Methodological Template for the Report

Use the following wording style when describing each test in a final report:

We use the cat-chair item as a controlled aligned example: the image branch receives pixels, while the text branch receives the sentence "A cat is sitting on a chair." The row identity is semantic rather than modality-identical. We then ask which invariants survive across encoders: local neighborhood topology, ranked distances, global CKA geometry, linearly shared subspaces, near-isometric Procrustes alignment, and directional retrieval after mapping. This separates the existence of shared structure from the stronger claim that one space can be exported into another at retrieval precision.

Final methodological summary

The combined methodology does not assume that all models share one literal embedding space. It tests a hierarchy: shared neighbors, shared ranking, shared global geometry, shared linear subspace, near-isometry, and exact retrieval after mapping. The shared Hub is then treated as an empirical object: its dimension is selected by sweep, its maps are evaluated directionally, and its failures are diagnosed through AUC, R@K, anisotropy, hubness, and effective rank. The practical question is not whether vectors look the same, but whether the structure they encode can be preserved, translated, or routed through a common space.